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Forecasting Coronal Mass Ejections Using Transformers: A Comparative Study of Flare-Associated Versus Flare-Independent Approaches

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Key Points:

- Forecasting coronal mass ejections (CMEs) using Space Weather Active Region Patches keywords is challenging even when using the evolution of these features over the past 24 hr
- Coupling CME forecasting to flare forecasting improves Matthews Correlation Coefficient but lowers Average Precision. Each model excels on different metrics
- Coronal data and better metrics that consider event-active ratios are needed for reliable CME forecasts

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Abstract

Forecasting coronal mass ejections (CMEs) remains a challenge, and no reliable, accurate forecasting method has yet been developed. Knowing that CMEs can occur in association with flares, we compare two approaches: (a) forecasting CMEs by coupling to flare forecasting, and (b) forecasting CMEs independently of flare occurrence. To do this, we train three transformer-based models that use sequences of HMI-SHARP keywords: (a) a flare model that predicts flares of class ≥M1.0, (b) a flare-CME model that predicts CMEs associated with ≥M1.0 flares, and (c) an independent CME model that predicts CMEs regardless of flare activity. When evaluated independently, the flare, flare-CME, and CME models achieve True Skill Score, Matthews Correlation Coefficient, and Average Precision (AP) of (0.79, 0.40, 0.46), (0.46, 0.27, 0.22), and (0.15, 0.16, 0.45), respectively. When the flare and flare-CME forecasts are chained and evaluated only during the 24 hr preceding a ≥M1.0 flare, the coupled model outperforms the stand-alone CME model with ~100% Bayesian confidence and a mean ΔMCC $\approx$ 0.18 (95% Highest Density Interval (HDI) [0.08, 0.27]). However, in terms of AP, the independent CME model outperforms the coupled model with ~100% Bayesian confidence and a mean ΔAP $\approx$ 0.15 (95% HDI [0.09, 0.21]). All metrics vary with the fraction of active regions that produce events, irrespective of class imbalance, so this ratio should be reported alongside the scores. We conclude that the community needs to agree on what constitutes a "best" model, and that further progress may require knowledge of the state of the solar corona.

Plain Language Summary

Coronal mass ejections (CMEs) are large eruptions of plasma and magnetic field from the Sun that can interact with the Earth's magnetic field, leading to geomagnetic storms that can disrupt and damage critical systems such as electrical grids. It is crucial to have a reliable forecasting system for these eruptions to mitigate their impacts. Despite previous work on forecasting them, a reliable forecasting system is still beyond reach. In our work, we study whether forecasting CMEs coupled with their associated flares, another highly energetic phenomenon in the Sun, is better or worse than forecasting the CMEs independently. We do this using machine learning methods that track changes in the Sun's surface magnetic field over 24 hr. We find that each approach performs differently depending on the performance measure used, and that forecasting CMEs remains challenging. We suggest that this may be because the magnetic field information used as input to the models does not contain enough information about the structures responsible for CME initiation, which may not be visible at the Sun's surface.

1. Introduction

Solar flares and coronal mass ejections (CMEs) are the most energetic phenomena occurring in the Sun's atmosphere. Although they represent different types of solar activity, with CMEs being eruptions of large quantities of plasma and magnetic field (Forbes, 2000), whilst flares produce intense electromagnetic radiation across the spectrum (Fletcher et al., 2011), both can result from a reconfiguration of the coronal magnetic field that is described by the CSHKP model (also called the standard model) of CMEs (Carmichael, 1964; Hirayama, 1974; Kopp & Pneuman, 1976; Sturrock, 1966). Studies of the association between flares and CMEs, such as Yashiro et al. (2005), find that, statistically, the larger the energy of a solar flare determined from its GOES class, the more likely it is to be associated with a CME. With *M* class flares having a 49% association rate and *X* class flares 91% (Yashiro et al., 2005).

Although no reliable and operationally implemented CME forecasting method has been developed yet, several signatures have been identified as indicators of future CME occurrence. For example, sigmoids, which are S-shaped structures observed in soft X-ray and EUV images, appear in regions with a higher likelihood of producing a CME (Canfield et al., 1999; Pevtsov et al., 1996; Rust & Kumar, 1996). However, sigmoids are not always visible due to projection effects, since they are better identified when looking from above rather than from the side. In addition to sigmoids, eruptive filaments routinely display a slow-rise phase—typically $1-15$ km s^{-1} lasting from tens of minutes to several hours—before their rapid acceleration and eruption (e.g., Sterling and Moore (2005); Shen et al. (2012); Cheng et al. (2020)). Hot flux ropes, defined as a diffuse active region-size emission in EUV bands corresponding to twisted magnetic field containing hot plasma ($\sim$10MK) (Cheng et al., 2011; Nindos et al., 2015; Reeves & Golub, 2011) have been found minutes before an eruptive flare (Cheng et al., 2013; Reeves & Golub, 2011; Zhang et al., 2012) or in confined flares to then erupt after tens of minutes or days (James et al., 2017; Kliem et al., 2021; Patsourakos et al., 2013). Regarding CME occurrence in active regions, flux cancellation is often observed beforehand and this may contribute to the formation of the eruptive structure and the accumulation of free magnetic energy (Amari et al., 2000; Linker et al., 2003; Wang et al., 2002). Newly emerging flux can also destabilize coronal fields and lead to eruptions by reconnecting with existing fields (e.g., Feynman and Martin (1995); Archontis and Hood (2012); Chen and Shibata (2000)), while shearing motions inject free magnetic energy and helicity into the magnetic field, indicating a possible future eruption (e.g., Romano et al. (2005); Jacobs et al. (2006); Vemareddy et al. (2012)). Active regions may also be classified according to the complexity of their photospheric magnetic field, as per Hale et al. (1919). Sammis et al. (2000) found that active regions classified as $\beta\gamma\delta$ are more likely to produce large flares. These active regions are also more likely to produce a CME, since larger flares are more likely to be associated with CMEs.

The photospheric vector magnetic field of active regions may also be used to derive quantities related to CME occurrence. The Space Weather Active Region Patches (SHARP) (Bobra et al., 2014) track active regions on the Sun, including both young emerging regions and old decaying regions, and calculate a set of parameters describing the photospheric vector magnetic field, as measured by the Heliospheric and Magnetic Image (Scherrer et al., 2011) onboard the Solar Dynamics Observatory (SDO) (Pesnell et al., 2011), every 12 min. These parameters are known as the SHARP keywords. Most works that use machine learning (ML) techniques to forecast solar activity (see below) make use of these parameters. They cover a wide range of quantities, such as the R value (Schrijver, 2007), which quantifies the magnetic flux along high-gradient polarity inversion lines and which was found to correlate with solar activity.

With the recent rise of ML, applications of these methods to the forecasting of flares and CMEs have appeared in the literature. For example, Abduallah et al. (2023) use a transformer-based model and sequences of SHARP keywords to forecast flares and their magnitude in the next 24, 48 or 72 hr. Similarly, Inceoglu et al. (2018) forecast solar eruptive events, including CMEs, flares and CMEs associated with a flare, using a support vector machine and a multilayer perceptron. While most works focus on flares, there have also been applications of these techniques that try to forecast a CME associated with a flare. Bobra and Ilonidis (2016) used SHARP keywords to predict whether a flare of GOES class $\geq M1.0$ will have an associated CME. Later, Liu et al. (2020) used sequences of SHARP keywords with a recurrent neural network to forecast whether an M- or X- GOES class flare will have an associated CME. While we have only named a few works, there is a missing connection in the literature between the forecasting models that predict whether a flare will have an associated CME and the flare forecast models on which they depend. That is, in order to predict whether a flare will have an associated CME, we must first predict whether a flare will occur, so these two forecast models are linked. Despite this, they are presented separately in the literature. Additionally, the forecast of CME occurrence independent of flares has not yet been tackled.

In this work, we compare two approaches to CME forecasting: forecasting whether a $\geq M1.0$ class flare will have an associated CME, the method used so far in the literature, versus forecasting a CME independently of any flare association. The latter has the advantage of potentially capturing a larger number of CMEs, since not all CMEs are flare-associated. Furthermore, coupling CME prediction to flare prediction introduces an additional failure point: any error in the flare model propagates to the CME forecast. Given that the relationship between flares and CMEs is definite but not one-to-one, decoupling the models offers a more robust and general forecasting strategy. To do so, we bridge the gap between the flare and CME forecast models by producing a flare forecast and then a flare-CME association forecast for the cases when a flare has been predicted. This differs from previous work in which the flare-CME association model was evaluated independently of the flare forecast model. We focus on forecasting, 24 hr in advance: flares of class $\geq M1.0$, whether these flares will have CMEs associated with them, and CMEs independently of their association with an $\geq M1.0$ class flare. We therefore train three models:

1. *Flare Forecast Model*: A model that forecasts whether an $\geq M1.0$ class flare will occur in a SHARP active region within the next 24 hr.
2. *Flare-CME Model*: A model that forecasts whether, given that an $\geq M1.0$ class flare will occur within 24 hr in a SHARP active region, a CME will be associated with that flare.
3. *CME Model*: A model that forecasts whether a CME will be produced in a SHARP active region within the next 24 hr independently of flare occurrence.

Our models are based on the transformer architecture and make use of temporal sequences of SHARP keywords. These parameters have been used extensively for similar forecasting tasks–see for example the works we described earlier. The transformer architecture was introduced by Vaswani et al. (2017a) and makes use of the concept of self-attention, which allows the model to focus on different parts of the input sequence when making predictions. Our model makes use of a sequence of magnetic parameters over the 24 hr prior to the forecast being issued to make a prediction for the occurrence of an flare/CME/flare + CME in the next 24 hr.

The goal of this work is not to provide a state-of-the-art forecast model. Rather, these models have been designed to explore how the flare and Flare-CME models interact when coupled, and the decoupling of CMEs from the forecast of flares.

2. Data

2.1. Data Sources

The main data used in this study consist of the SHARP keywords, which serve as features in our models to produce forecasts. Additionally, a flare catalog and a CME catalog are used to label active regions based on whether they produce one of these events within 24 hr after the forecast is issued. The data sources are described in detail below.

SHARP Keywords: A set of parameters describing the vector magnetic field of SHARP active regions every 12 min (Bobra et al., 2014). They describe different aspects of the magnetic field, including values such as the total unsigned magnetic flux and the mean shear angle of the magnetic field. Although available from the Joint Science Operations Center of the SDO, we make use of the curated SHARP data from Angryk et al. (2020), which is ready to be used for training ML models. We only use data entries with the "is trusted magnetic field" column set to 1 (see section *SHARP data and magnetic field parameters* from Angryk et al. (2020)) and with no missing values for the keywords. The data released by Angryk et al. (2020) is available from 2010 to the end of 2018, limiting us to this period. We make use of all the keywords described in Table 1 of Angryk et al. (2020). Additionally, we calculate event history parameters as described in Section 2 of Liu et al. (2019), extending them to include CME history.

Flare catalog: Including the location, intensity and time of flares along with the region that produced them. This is also obtained from Angryk et al. (2020) and the main difference from the official GOES catalog is that cross-checking with SSW Latest Events and Hinode-XRT is performed to ensure the correctness of the flare records.

CME catalog: Relating CMEs from the LASCO CDAW catalog (this CME catalog is generated and maintained at the CDAW Data Center by NASA and The Catholic University of America in cooperation with the Naval Research Laboratory. SOHO is a project of international cooperation between ESA and NASA. https://cdaw.gsfc.nasa.gov/CME_list/) to their source SHARP active region. To obtain these associations we make use of post-eruptive signatures that point to the origin of a CME. In particular, dimmings and flares as described in Hernandez Camero et al. (2025).

2.2. Data Preparation and Processing

The data sets we create for our study all follow the same structure; given an observation period of 24 hr during which a sequence of features is collected, a binary label is assigned to the data depending on whether an event will occur in the forecast period, which spans the 24 hr after the end of the observation period. This setup follows prior work (Bobra & Ilonidis, 2016; Liu et al., 2019, 2020) and reflects a timescale that is both operationally useful and practically learnable. While transformer models could, in principle, use longer or variable observation periods,

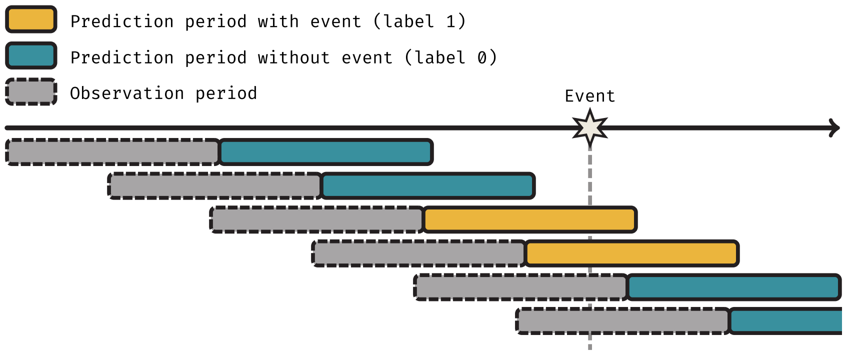

Figure 1. Simplified process for the creation of the data sets. Given the timeline for a particular Space Weather Active Region Patches active region, for each step (row in this figure), an observation period (dashed) and prediction period (solid) are considered. If an event occurs during the prediction period, the data associated with the observation period is labeled with 1 and 0 otherwise. Then, for the next step the periods are shifted by some amount and the process repeated until the end of the region's lifetime is reached. This is repeated for all regions of the data set.

fixing it to 24 hr ensures consistency across samples, simplifies training, and allows for direct comparison with existing studies.

For the flare model, an event refers to the region of interest producing a $\geq M1.0$ class flare, for the flare-CME model it refers to, *assuming the region will produce an* $\geq M1.0$ *class flare*, the flare having an associated CME and for the CME model it refers to the region of interest producing a CME irrespective of any flare association. Figure 1 shows schematically how the process of creating the data sets works, which we describe below:

1. Begin by selecting one SHARP active region (identified by its HARPNUM). Starting from the first data point available, use the next 24 hr as the observation period from which data will be obtained and the 24 hr after the end of the observation as the forecasting period, from which the label will be obtained.
2. Ensure the validity of the data during the observation period. This means that at least 80% of the data points must be valid, that is 80% of the records should have the "is trusted magnetic field" flag set to 1 and no missing values. Additionally, the last point of the sequence must be valid. If this is not the case, move directly to step 4 and reject this data sequence.
3. Obtain a label for this period. In the case of flares and independent CMEs this is simply whether there's an event (flare of class $\geq M1.0$ or CME respectively) in the next 24 hr. In the case of the flare-CME data set, first check if there's a flare of class $\geq M1.0$ in the next 24 hr. If there is not, this data is not used as part of this data set and we skip to the next step. Otherwise, the label is whether a CME was associated with the flare.
4. Move the observation and forecast periods by 12 min. Repeat from step 2 until the end of the region's lifetime.

While it would be possible to require no events in the observation period (i.e., no flares or CMEs), this reduces the amount of data significantly. This is because regions that produce events tend to produce more than one, so overlaps are common. For these reasons, we do not impose this condition.

In ML studies, it is then common to split data sets into training, validation, and test sets. The training set is used to train the model, which is then evaluated on the validation set during training to monitor its performance. This helps detect overfitting—a scenario where the model "memorizes" the training data, performing exceptionally well on it but poorly on unseen data. To mitigate overfitting, the validation set provides an unbiased estimate of the model's generalization ability during training. However, hyperparameters, which are manually adjusted parameters not directly learned during training, can also become overfitted to the validation data. To address this, a separate test set is used to evaluate the model only after all hyperparameters are finalized and the model is fully trained.

Given the limited number of events in our data set (467 $\geq M1.0$ class flares, 600 CMEs and 135 CMEs associated with a $\geq M1.0$ class flare), splitting the data into three distinct sets (training, validation, and test) is not feasible. This issue is particularly problematic in the case of associating CMEs with flares, where the number of events is especially low (135). To overcome this limitation, we adopt a form of k-fold cross-validation, where the data set is divided into 10 splits (chapter 7.10.1, Hastie et al. (2009)). Each split is created by grouping all data belonging to

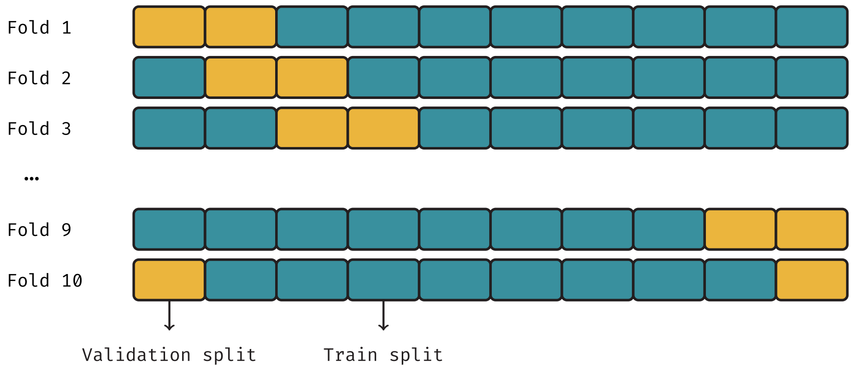

Figure 2. Cross-validation strategy followed in this work. The whole data set is divided into 10 splits (each rectangle representing a split) and for each fold we train a model using 8 splits as training data and 2 splits as validation data. Then, we evaluate metrics on all folds and report the means and standard deviations over the 10 independent models.

specific SHARP active regions, ensuring there is no data leakage between splits. This means, for each forecasting problem (CME, Flare and Flare-CME models) there will be 10 realizations of each model, each using 8 splits for training and 2 for validation. In this way, each realization uses 80% of the data for training and 20% for validation. This process is illustrated schematically in Figure 2, where each fold represents an independent realization of the model. The final performance is presented as the mean and standard deviation across these 10 realizations.

To ensure fair comparison between models, these splits are consistent across the three data sets. While it is possible to use methods like random shuffling followed by splitting, this approach has significant drawbacks. Data points are not truly independent, as they can overlap in time (as shown in Figure 1, where observation periods overlap), and even if they do not, they may still belong to the same SHARP active region. One alternative is to split the data by year, minimizing temporal overlap; however, this makes it challenging to ensure balanced splits in terms of event numbers and could bias the model toward certain phases of the solar cycle. For these reasons, we choose to split the data by active regions, ensuring that all data from any given SHARP active region falls within a single split.

Distributing the regions across the 10 splits to evenly balance the CMEs, flares, and their associations is a complex optimization problem. We used the PulP Python library for linear programming to tackle this. Given the uncertainty around solving such an optimization, we allowed it to run for 10 min, which resulted in a reasonably good, though not perfect, distribution of events across the splits, as shown in Table 1.

The inputs to the model are sequences of 40-dimensional vectors (corresponding to SHARP keywords), with shape $(60, 40)$, representing the evolution of each active region over the past 24 hr at a 24-min cadence. Prior to training, each feature (keyword) is normalized using z-score normalization:

$$z_i = \frac{x_i - \mu_{i,\text{train}}}{\sigma_{i,\text{train}}} \tag{1}$$

where x_i is the raw value of the ith feature, z_i is the corresponding normalized value, and $\mu_{i,\text{train}}, \sigma_{i,\text{train}}$ are the mean and standard deviation of that feature computed only on the training set. These statistics are then used to normalize both the training and validation data to prevent data leakage.

3. Model Architecture, Training, and Evaluation

3.1. Model Architecture

Figure 3 shows the architecture used in our models, which consists of 2 transformer blocks followed by a fully connected classification layer. For details of the implementation, we provide details of our architecture in Appendix A.

Table 1
Summary of the Folds Used in the Cross-Validation Training of the Models

Fold	Type	# SHARP	# ≥M1.0 flares	# CME	# CMEs with ≥M1.0 flare
0	Val.	573	84 (0.02) [0.06]	146 (0.04) [0.14]	22 (0.30) [0.43]
	Train	2308	383 (0.02) [0.05]	619 (0.04) [0.11]	113 (0.35) [0.48]
1	Val.	587	83 (0.02) [0.06]	146 (0.04) [0.15]	24 (0.31) [0.52]
	Train	2294	384 (0.02) [0.05]	619 (0.04) [0.11]	111 (0.35) [0.45]
2	Val.	628	91 (0.02) [0.06]	146 (0.04) [0.12]	25 (0.35) [0.56]
	Train	2253	376 (0.02) [0.05]	619 (0.04) [0.12]	110 (0.34) [0.44]
3	Val.	602	92 (0.02) [0.06]	151 (0.04) [0.12]	27 (0.35) [0.53]
	Train	2279	375 (0.02) [0.05]	614 (0.04) [0.12]	108 (0.34) [0.45]
4	Val.	567	92 (0.02) [0.06]	156 (0.04) [0.12]	28 (0.32) [0.41]
	Train	2314	375 (0.02) [0.05]	609 (0.04) [0.12]	107 (0.35) [0.48]
5	Val.	580	92 (0.02) [0.04]	156 (0.04) [0.10]	28 (0.33) [0.38]
	Train	2301	375 (0.02) [0.05]	609 (0.04) [0.12]	107 (0.35) [0.48]
6	Val.	574	100 (0.02) [0.03]	160 (0.05) [0.09]	28 (0.35) [0.56]
	Train	2307	367 (0.02) [0.06]	605 (0.04) [0.13]	107 (0.34) [0.45]
7	Val.	558	100 (0.02) [0.04]	160 (0.05) [0.10]	30 (0.41) [0.55]
	Train	2323	367 (0.02) [0.06]	605 (0.04) [0.12]	105 (0.32) [0.45]
8	Val.	539	100 (0.02) [0.05]	157 (0.04) [0.13]	32 (0.39) [0.41]
	Train	2342	367 (0.02) [0.05]	608 (0.04) [0.12]	103 (0.33) [0.48]
9	Val.	272	54 (0.02) [0.05]	79 (0.04) [0.13]	16 (0.35) [0.38]
	Train	2609	413 (0.02) [0.05]	686 (0.04) [0.12]	119 (0.34) [0.47]

Note. Values in brackets represent the class imbalance and values in square brackets the event-active ratio which is the number of SHARP regions that produce at least one event divided by the total number of SHARP regions in the data set.

3.2. Model Training

We train our models using the Adam optimizer and a cyclic learning rate scheduler. All model and data set configurations are available in Hernandez Camero (2025b).

3.3. Model Evaluation

After training the models for each of the three forecasting scenarios, we evaluated them using a number of metrics. The selection of metrics is based on the most commonly used metrics in the literature. It should be noted that no single metric presents a completely unbiased evaluation of a model's performance. Each metric has its own strengths and biases, and we caution against relying on a single metric to evaluate a forecasting model. Instead, we find that inspecting a large number of model predictions visually compliments the metrics by providing a better overview of the model behavior (cf. Figure 8).

Our model architecture outputs a continuous number between 0 and 1 which is transformed into binary labels for no event (0) and for event (1) in the next 24 hr by defining a threshold. While the selection of this threshold influences metrics and could be optimized to enhance specific scores, choosing such an "optimal" threshold involves operational considerations and trade-offs among metrics, and any threshold selected directly from test data would introduce bias. Given these complexities, we follow Leka et al. (2019, 2023) and adopt a fixed threshold of 0.5 for all models. Once predictions are thresholded, a confusion matrix is defined, whose entries are:

True Positive (TP) The number of positive examples that were correctly classified as positive. For example there is a CME in the next 24 hr (target 1) and we predict there will be one (prediction 1).

True Negative (TN) The number of negative examples that were correctly classified as negative. For example there is no CME in the next 24 hr (target 0) and we predict there will be none (prediction 0).

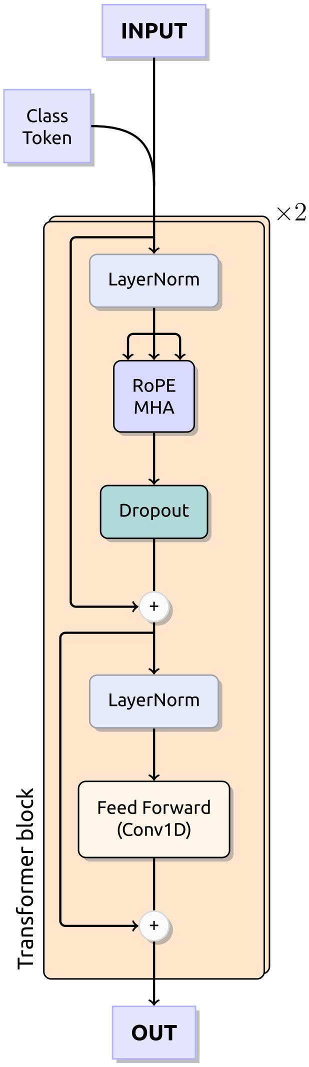

Figure 3. Architecture of the models used in this work. A class token is appended to the input sequence before being processed to serve as its global representation. Each transformer block starts by applying a layer normalization (LayerNorm, Ba et al. (2016), which normalizes across features of each instance) to the input, followed by a Multi-Head Attention (MHA, Vaswani et al. (2017b)) mechanism with Rotary Position Embeddings (rotary positional encoder (RoPE), Su et al. (2021)) with dropout to prevent overfitting. The output of these operations is added to the input sequence using a residual connection before continuing. Next, layernorm is applied and followed by a feed-forward network using 1D convolutions. The final output is added to the output of the RoPE MHA part through another residual connection. The model makes use of two of these transformer blocks connected sequentially using the processed class token to produce the final production using a single fully connected layer with 1 neuron.

False Positive (FP) The number of negative examples that were incorrectly classified as positive. For example there is no CME in the next 24 hr (target 0) but we predict there will be one (prediction 1).

False Negative (FN) The number of positive examples that were incorrectly classified as negative. For example there is a CME in the next 24 hr (target 1) but we predict there will be none (prediction 0).

Using these we can define our metrics. The True Skill Score (TSS) is calculated as

$$\mathrm{TSS} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}} - \frac{\mathrm{FP}}{\mathrm{FP}+\mathrm{TN}} \tag{2}$$

and it is a balance of the TP rate (TPR) and the false positive rate (FPR). The TPR is the proportion of positive examples that were correctly classified as positive and the FPR is the proportion of negative examples that were incorrectly classified as positive. The TSS ranges from −1 to 1, with 1 being the best possible score and 0 being the score of a random classifier. Negative values indicate something is wrong with the model as it is classifying data opposite to what it should. We also calculate the Matthews Correlation Coefficient (MCC) (Matthews, 1975):

$$\mathrm{MCC} = \frac{\mathrm{TP}\times\mathrm{TN} - \mathrm{FN}\times\mathrm{FP}}{\sqrt{(TP+FN)(\mathrm{TN}+\mathrm{FN})(TN+FP)(\mathrm{TP}+\mathrm{FP})}} \tag{3}$$

which ranges from −1 to 1 with 0 a random model and 1 the best performance and improves upon shortcomings of the TSS by being more stable to changes in data set composition (Francisco et al., 2025). Finally, we also calculate the average precision score (AP) (Zhang & Zhang, 2009) which is defined as the area under the Precision-Recall (PR) curve. This curve shows how the precision of the model (how many of the predicted positives are actually positive) changes with the recall (how many of the true positives did we correctly classify as positive) by varying the threshold used to convert the continuous predictions into binary. The AP ranges from 1 to the proportion of positive examples in the data set. A random classifier would have an AP equal to the proportion of positive examples in the data set. A perfect classifier would have an AP of 1. The AP is a good metric to evaluate the model when the class imbalance is large, as it is the case in our data sets (cf. Table 1) and it does not depend on the value of a threshold.

4. Results

4.1. Individual Results

Table 2 summarizes the metrics for each model. The models which forecast whether a flare will have an associated CME (flare-CME model) or whether a CME will happen regardless of flare association (CME model) achieve lower TSS values, 0.13 and 0.52 and MCC values 0.27 and 0.16 respectively than the model that predicts whether an $\geq M1.0$ class flare will occur, which achieves a TSS of 0.82 and an MCC of 0.4. However, though the flare-CME model has by far the smallest TSS, it achieves a similar AP to the flare forecast, 0.45 and 0.46 respectively, while the CME alone model has the lowest at 0.22. This hints at the difficulty of evaluating a model simply through the use of one metric.

Table 2
Validation Metrics for Each Forecasting Tasking Task Independently

Forecast type	Threshold	TSS	MCC	AP
Flare	0.50	0.79 (0.06)	0.40 (0.05)	0.46 (0.12)
CME	0.50	0.46 (0.07)	0.27 (0.04)	0.22 (0.06)
Flare-CME	0.50	0.15 (0.10)	0.16 (0.10)	0.45 (0.08)

4.2. Comparison of CME Forecast Models

To compare the two approaches to forecasting CMEs, we must evaluate both models on the exact same data set. The flare-CME model predicts whether a flare of class $\geq M1.0$ will be associated with a CME, so we evaluate both models on the flare-CME data set, which encompasses only the 24 hr prior to a $\geq M1.0$ class flare.

For the flare-CME model, we first generate a prediction using the flare model. If an $\geq M1.0$ class flare is predicted within the next 24 hr, the flare-CME model then predicts whether that flare will be associated with a CME. If no flare is predicted, we conclude that no CME will occur. This is the coupled Flare and Flare-CME model (Flare + Flare-CME). On the other hand, the CME model directly predicts whether a CME will occur during the same 24 hr period, independent of any flare prediction. This means the CME model runs regardless of whether the flare model's prediction is correct or not. After running both models on the same validation data set, we present the results in Table 3. This evaluation allows us to realistically assess the performance of the flare-CME model, which depends on an accurate flare forecast, compared to the independent CME model.

In order to statistically asses if either of the model is outperforming the other we performed a paired analysis of the MCC and AP scores. Defining $\Delta_i MCC = \mathrm{MCC}_{\text{Flare-CME}_i} - \mathrm{MCC}_{\text{CME}_i}$ as the difference in MCC score between the Flare + Flare-CME model and the CME model for cross-validation fold i, the vector $\boldsymbol{\Delta} MCC$ (of size 10; see Figure 2) contains the paired differences across all folds. We analyzed these differences using Bayesian estimation of the paired mean difference under a Student-t likelihood, following the framework of Kruschke (2013). This approach models the distribution of $\Delta_i MCC$ with a Student-t likelihood and infers a posterior distribution for the mean difference $\mu_{\Delta MCC}$. A value $\mu_{\Delta MCC} > 0$ indicates that the Flare + Flare-CME model outperforms the CME model according to MCC at threshold 0.5, and vice versa.

Our analysis yielded $P\left(\mu_{\Delta MCC} > 0 \,|\, \boldsymbol{\Delta} MCC\right) \sim 1$, with a 95% Highest Density Interval (HDI) of $[0.0812, 0.2700]$. We therefore conclude that, *during periods of intense* $\geq M1.0$ *class flare activity*, forecasting CMEs via a model that treats them as consequences of flares is likely to perform better than forecasting them independently according to the MCC metric with a threshold of 0.5, though the magnitude of the improvement is modest.

Similarly, with $\Delta_i AP = \mathrm{AP}_{\text{Flare-CME}_i} - \mathrm{AP}_{\text{CME}_i}$ and $\boldsymbol{\Delta} AP$ we find $P\left(\mu_{\Delta AP} > 0 \,|\, \boldsymbol{\Delta} AP\right) \sim 0$, with a 95% HDI of $[-0.2052, 0.0895]$. Therefore, according to the AP score, the situation is the reverse and the CME model outperforms. For completeness, we also show the precision-recall curves of the models in Figure 4.

5. Discussion

5.1. The Challenge of Forecasting CMEs

The results in Table 2 and the CME forecast comparisons in Table 3 highlight that forecasting CMEs, whether associated with a flare or not, is more challenging for our models than predicting $\geq M1.0$ class flares. This discrepancy may arise from the physical nature of CMEs, which result from a loss of magnetic equilibrium in the solar corona (for a review see Green et al. (2018)). This means that a configuration or evolution of the coronal magnetic field that is likely to erupt may not leave clear signatures in the photosphere, where the SHARP-derived magnetic parameters used in our models are measured. Incorporating coronal information, such as images from the Atmospheric Imaging Assembly onboard SDO (Lemen et al., 2011), could help test this hypothesis. However, no equivalent to the SHARP keywords is currently available for coronal data, and direct measurements of coronal magnetic fields remain difficult.

Table 3
Validation Results for Comparison of the Forecasting of CMEs Independently of Flares (CME) and CMEs Associated With $\geq M1.0$ Class Flares Using the a Threshold 0.5 for Both Models

Forecast type	TSS	MCC	AP
CME	−0.01 (0.05)	−0.02 (0.08)	0.56 (0.12)
Flare + Flare-CME	0.15 (0.10)	0.16 (0.11)	0.41 (0.12)

Note. The models are evaluated only on the Flare-CME validation data set, which encompasses the 24 hr before every $\geq M1.0$ class flare. The Flare + Flare-CME model is coupled so that if no flare is forecasted, then the Flare-CME prediction is taken to be 0. Otherwise, we use directly the Flare-CME prediction.

Statistical tests on MCC and AP show no single best model; each excels on a different notion of skill. These metrics also have issues associated with them, some of which are discussed below, which make declaring one model better than another even more difficult. The community has yet to agree on a testing framework or on which metric should define "best." Therefore, under the contradiction of the two metrics, we refrain from identifying a single best

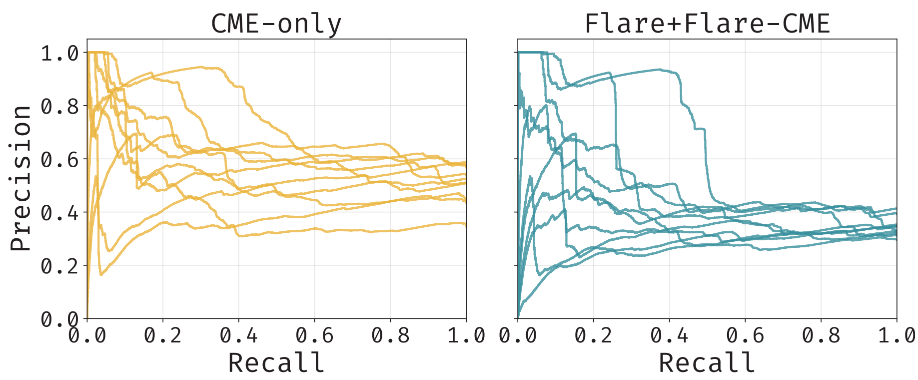

Figure 4. Precision-Recall curves for the coronal mass ejection (CME) and combined Flare + Flare-CME model. Each line represents an individual fold of the model. While both models show similar tendencies, the CME model generally has higher precision values for the same recall values. This explains its higher Average Precision score.

model. Nevertheless, the two models do show different behaviors as shown in the metrics which may be explained by the nature of the forecasting problem each has been trained to solve. The CME and flare models are each trained to identify whether a region's magnetic configuration is likely to produce a CME or a flare, respectively. These models are exposed to extremes of relatively simple and relatively complex magnetic field configurations. In contrast, the flare-CME model is trained to distinguish between large flares with and without associated CMEs. Because this model is only exposed to regions that produce a $\geq M1.0$ class flare, it is effectively conditioned on strong magnetic signatures. As a result, we speculate it may be forced to learn subtler features that distinguish between eruptive and non-eruptive flares. We speculate that the CME model may not rely so much on such patterns.

Testing these interpretations would require the application of model explainability techniques, which is beyond the scope of this work. Our focus is on comparative model performance, not on interpreting internal mechanisms. Moreover, given the early and still-evolving nature of explainability in deep learning, particularly for time-series transformer architectures, we prefer not to introduce potentially misleading or unvalidated attributions. We believe meaningful progress in this area will require dedicated, physically grounded interpretability methods, which we leave for future work.

5.2. The Impact of Event Distributions on the Evaluation Metrics

We found that the way CMEs and flares are distributed across the SHARP active regions in our data set significantly affects the interpretation metrics achieved by the models. To illustrate this, consider two types of data sets:

In the first data set, events (CMEs or flares) are concentrated in only a few regions (cf. left panel of Figure 5). Most regions have no events, which we call non-event-active regions, while those that produce at least one event are called event-active regions. In this type of data set, a model can achieve higher metric values simply by classifying regions into event-active (always predicting an event) or non-event-active (never predicting an event). We hypothesize this is because all of the metrics currently used in the literature fail to account for the non-independent nature of the data points which are derived from the same active region, effectively clustering data points into groups that may be easy to identify by the model. Meanwhile, identifying when a region is about to become active requires distinguishing more complex signatures within a region's data points. We refer to this as a baseline model that knows which regions will produce events but has no information about when they will happen.

In contrast, in the second data set, the same number of events is more evenly distributed across regions (see the right panel of Figure 5). Here, most regions produce at least one event, and the baseline forecast achieves lower metrics. This can be summarized using what we call the event-active ratio: the number of event-active regions divided by the total number of regions (see Table 1 for the values in our data sets).

Previous works such as Barnes et al. (2016); Leka et al. (2019, 2023) already noted how a higher TSS value may be achieved by an overforecasting system with low event rate (class imbalance). We believe that although the TSS

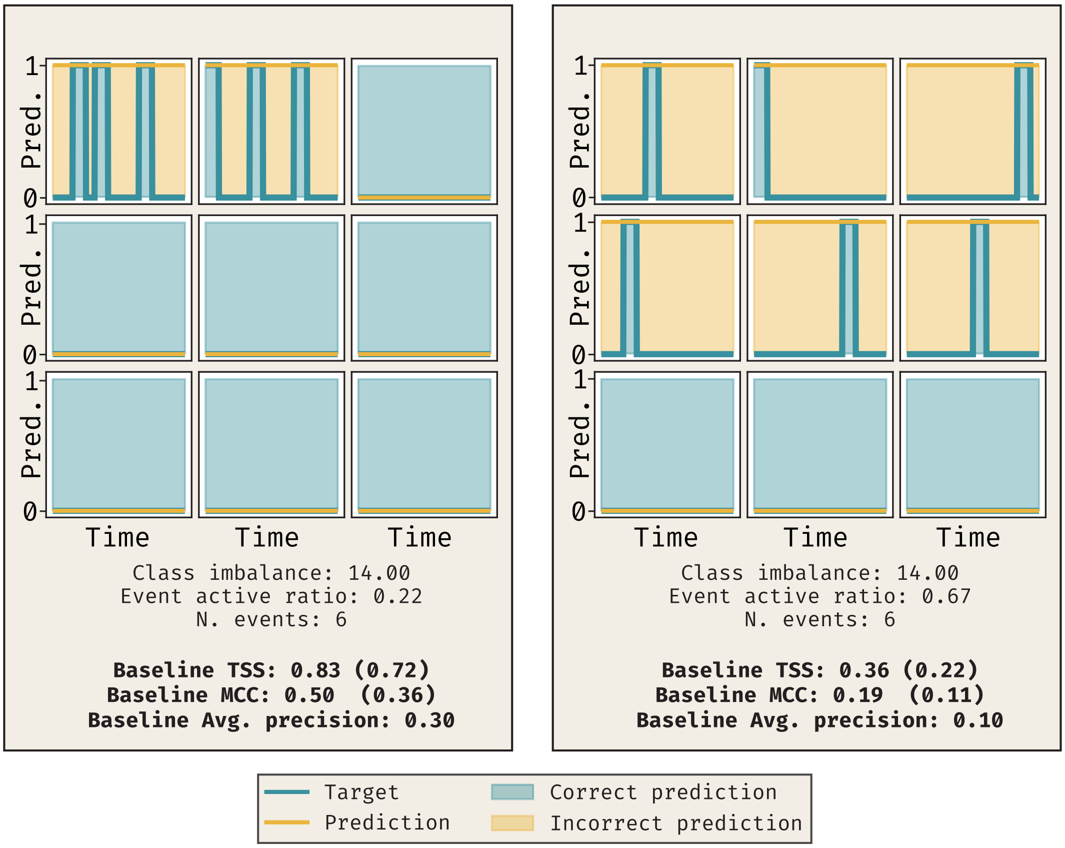

Figure 5. The ratio of event-active regions in a data set has an important effect on the interpretation of the metrics achieved by a model. Here we show a baseline forecast that can only distinguish between event-active and non-event-active regions. This baseline forecast predicts always 1 for event-active regions and always 0 for non-event-active. We show here two data sets consisting of 9 regions with 6 events. On the left panel, the 6 events are distributed along 2 regions, resulting in a small event-active ratio and a high baseline metrics. On the right panel, the 6 events are distributed along 6 regions, resulting in a higher event-active ratio and a significantly lower baseline metrics. Values in brackets are calculated using the weighted confusion matrix of Guastavino et al. (2024). Both panels have the same class imbalance ratio.

is independent of the class imbalance, their findings are a result of the low event-active ratios found for highly energetic flares (cf. Table 1, square brackets). Guastavino et al. (2024) and Francisco et al. (2025) also note how changes in behavior (when a region goes from being quiet to event-active and vice versa) are crucial in interpreting metrics. Models that perform badly during behavior changes can still achieve high scores in unbalanced settings. However, we find that under the same class imbalance and the same number of behavior changes in the two situations depicted in Figure 5, all metrics still show a higher baseline value in the data set with a lower event-active ratio. It is possible that metrics such as the weighted TSS introduced by Guastavino et al. (2024) could help in this case. The value in parentheses are calculated using the weighted approach and lower the metrics in both cases. Applying the weighted approach to our data sets was not possible due to limitations in data gaps, so the effect in our data is unknown.

To assess whether our models are sensitive to the even-active ratio, we:

1. Calculated the event-active ratios in the data sets for each of the three forecast models, as shown in the square brackets of Table 1. The flare-CME data sets have the highest ratio at ~0.38–0.56, which is expected since $\geq M1.0$ class flares are often associated with CMEs, while the flare (~0.03–0.06) and CME (~0.10–0.15) data sets have much lower ratios.
2. Calculated the metrics for each forecast model using a baseline model, as shown in Table 4. For each region that produced at least one event during its lifetime, we always predicted an event in the next 24 hr; for regions with no events, we always predicted no event. The flare ($TSS_{baseline} = 0.93$) and CME ($TSS_{baseline} = 0.81$)

Table 4
Validation Metrics for Each Forecasting Task Using the Baseline Forecasts

Forecast type	TSS	MCC	AP
Flare	0.93 (0.02)	0.47 (0.06)	0.24 (0.06)
CME	0.81 (0.02)	0.38 (0.02)	0.18 (0.02)
Flare-CME	0.55 (0.12)	0.54 (0.09)	0.54 (0.07)

models had the highest baseline TSS values, while the flare-CME model had the lowest ($TSS_{baseline} = 0.55$). This is consistent with our earlier discussion and suggests that the models might only be identifying which regions tend to produce events, without accurately predicting their timing.

3. Evaluated how well the models predict *when* an event will happen versus simply identifying *where* an event is likely. We progressively removed non-event-active regions from the evaluation data and analyzed the resulting metric values (Figure 6). As non-event-active regions were removed, the event-active ratio increased, and TSS values decreased noticeably. The MCC values showed a decrease as well but of smaller magnitude, while the AP increased slightly and then settled. This indicates that all these metrics are sensitive to changes in the underlying event-active ratio although with different behaviors.
4. Analyzed the distribution of the continuous predictions made by each model, as shown in Figure 7. In the top row, we included all regions in the validation data, while in the bottom row we included only event-active regions (corresponding to an event-active ratio of one in Figure 6). We separated predictions based on whether they corresponded to a true label of 1 (an event occurred in the next 24 hr) or 0 (no event occurred). Ideally, a perfect model would produce two distinct distributions with minimal overlap. However, once non-event-active regions were removed, the overlap increased. This suggests that the models tend to predict higher values for event-active regions, even when no event occurs, meaning they may be simply identifying regions prone to activity without predicting specific events accurately.
5. Examined the actual forecasts for specific regions, as shown in Figure 8. For some regions, like HARPNUM 2696, the model always predicted above the threshold, suggesting a CME regardless of the actual timing. In contrast, for other regions like 5,982, the model correctly predicted a CME only when it actually occurred. Meanwhile, for regioin 5,290, that never produced a CME, the model correctly stays under the threshold at all times. This indicates that while the models can identify regions likely to be active, they may struggle to determine the exact timing of these events.

6. Conclusion

We have trained three transformer-based models to forecast three scenarios 24 hr in advance: $\geq M1.0$ class flares (our flare model), CMEs associated with $\geq M1.0$ class flares (our flare-CME model), and CMEs independent of flare associations (our CME model), focusing on events originating in SHARP active regions. We evaluated the flare-CME model by considering both the predictions made by the flare model and the flare-CME model itself. If no flare is predicted by the flare model, we take the prediction of the flare-CME model to be negative (no CME). This is in contrast to previous works that focus on one model at a time.

We found that which approach (flare coupled vs. independent) is better depends on the metric of choice, and lacking a community accepted metric as the definition of "best" do not choose one over the other. However, CME forecasting shows consistently lower scores than flare forecasting, highlighting persistent challenges, perhaps linked to the lack of coronal data in the inputs.

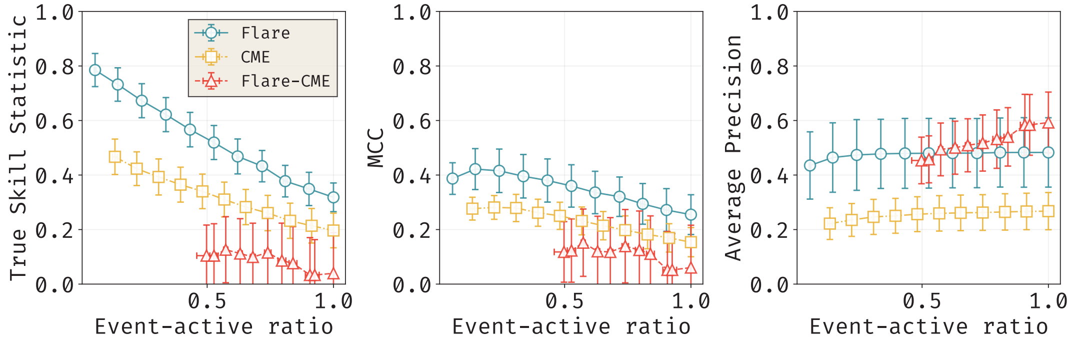

Figure 6. Plots showing the effect of evaluating models individually only on event-active regions on the True Skill Score (left), Matthews Correlation Coefficient (middle) and Average Precision (right) metrics. The event-active region ratio increases and the metrics change, showing different sensitivities to the change in ratio. The markers are the median value from the 10 folds of each model and the vertical bars the standard deviation.

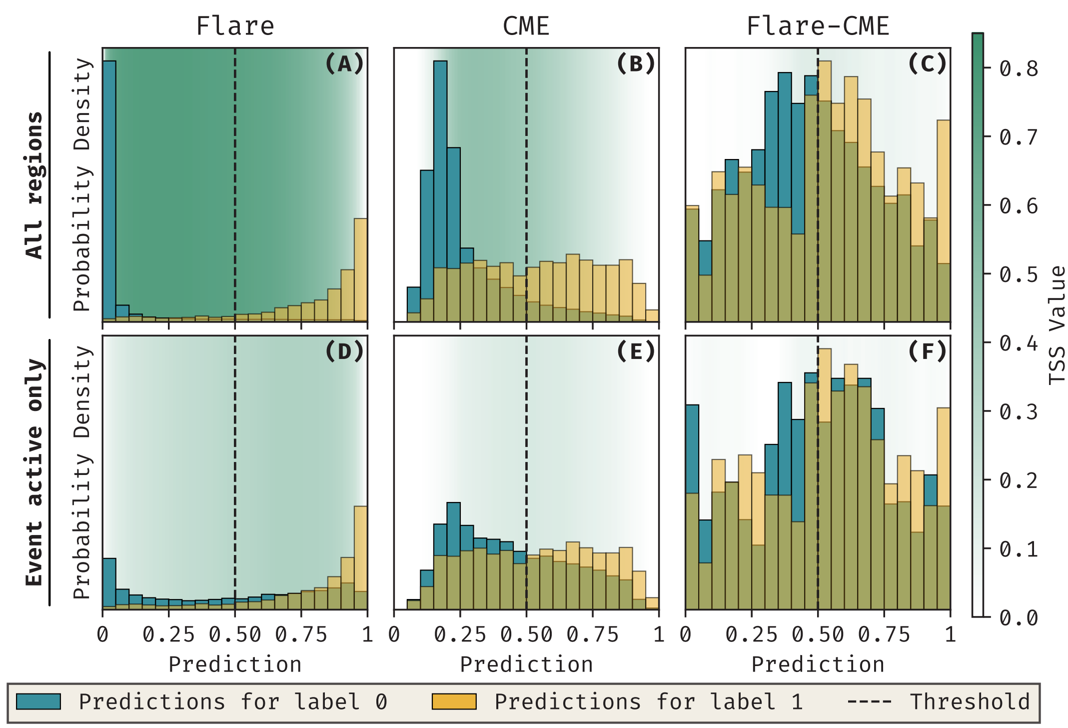

Figure 7. Distribution of predictions for each model. Predictions are grouped by their true label (0 or 1). Perfect models would show two clearly distinct distributions with minimal overlap. When evaluating only event-active regions, the overlap between distributions increases, indicating that models struggle to precisely forecast when events will happen.

We also discuss the implications of the non-independent nature of data derived from SHARP active regions for the interpretation of metrics. Models that classify regions into event-active (producing at least a single event) versus non-event-active (never producing an event) achieve unexpectedly large scores when the events are concentrated in a small number of regions. This highlights the need for metrics which account for the temporal aspect of the data in order to truly evaluate whether models know *when* an event is going to happen and not just *which* region is going to produce it.

In future work, we plan to focus on the explainability of these models to better understand which signatures they are identifying. This will not only help reveal the features and patterns in the data that contribute most to the predictions, but also provide a way to test our speculations about why the Flare + Flare-CME model performs

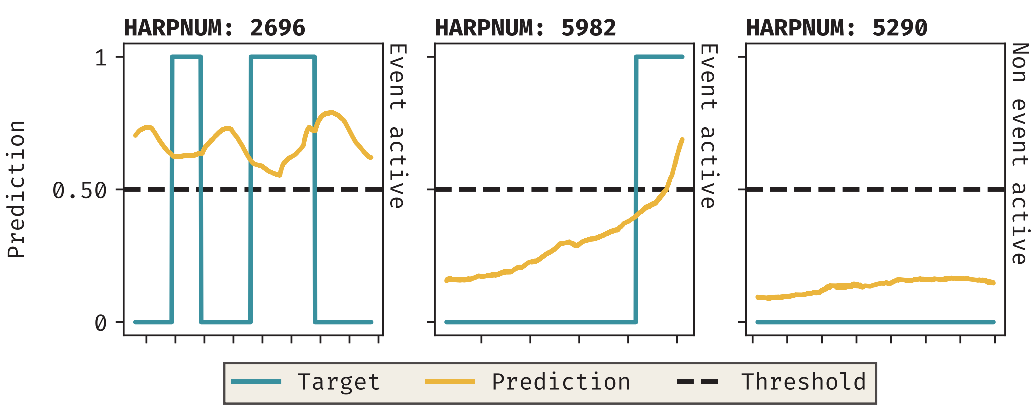

Figure 8. Coronal mass ejection forecast predictions for three Space Weather Active Region Patches active regions. Regions 2,696 and 5,982 are event-active, while region 5,290 is not. The model consistently forecasts above the threshold for region 2,696, failing to predict the exact timing of events. For region 5,982, the forecast only exceeds the threshold when an event is imminent. Predictions for region 5,290 remain consistently below the threshold. Each tick represents a 24 hr period.

differently to the independent CME model in the 24 hr prior to a $\geq M1.0$ class flare. By doing so, we can gain valuable insights into the model's decision-making process and improve our understanding of the underlying solar physics.

Appendix A: Details of Model Architecture

Our model makes use of a "class token." This is simply a learnable vector of shape equal to the number of features that is appended to the sequence of features. Its purpose is to serve as a global representation of the input sequence. It can be used at the end of the transformer blocks to generate a prediction. An alternative approach would be to use average pooling after the transformer blocks, averaging the output across the temporal dimension. Either way, a vector of shape equal to the number of input features is used to make the predictions.

We also make use of a rotary positional encoder (RoPE) (Su et al., 2024) in the first transformer block. The architecture of transformers is such that by default the model has no knowledge of the sequential nature of the data. The results would be the same whether the input data is in the right order or not. In order to make the model aware of the temporal nature of our data, positional encoders are used. Here we choose to use RoPE, although other options exist.

In our transformer block, the input is first processed by the multi-head attention mechanism, followed by a series of feed-forward layers. In our implementation, these are 1D convolutional layers, which are functionally equivalent to fully connected layers. Residual connections are used throughout the block to allow information to bypass certain layers, improving gradient flow and overall model performance.

Normalization is applied at two points within each block to improve training stability and help prevent overfitting. Specifically, we use Layer Normalization Ba et al. (2016), which normalizes each input across its feature dimension, ensuring the output has zero mean and unit variance. Unlike dataset-wide normalization (e.g., z-score), LayerNorm operates on individual data instances.

A search in hyperparameter space was performed to determine the best values. No significant changes to the metrics or model behavior was found and the hyperparameters that maximized the training TSS were chosen. We provide the full training logs of the final models in the experiment tracking platform Weights And Biases with links available in Hernandez Camero (2025a). The logs contain the full list of hyperparameters used in the runs, which can also be found in the config/folder in the source code.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Data Availability Statement

The model weights and training data used in the study are available at Zenodo via Hernandez Camero (2025a) with a MIT license. Version number 1.0.2 of the sourcecode was used for definition and training of the models and is preserved at Hernandez Camero (2025b), available via GNU license and developed openly at https://github.com/JulioHC00/cme%5Fflare%5Fforecast.

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This work made use of the following packages: Pytorch (Paszke et al., 2019), Astropy (Astropy Collaboration et al., 2013, 2018, 2022), Numpy (Harris et al., 2020), Pandas (pandas development team, 2020; Wes McKinney, 2010), Matplotlib (Hunter, 2007). We also acknowledge the following facilities: SDO (HMI), SOHO(LASCO), GOES. J.H.C. acknowledges the support of the Science and Technology Facilities Council (STFC) under grant number ST/X508858/1 for this work.

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